Empirical Angle Estimation for Historical Artillery under Aerodynamic Constraints

Team 155

Problem B

Abstract

This paper investigates the motion of a projectile during flight from an aerodynamic perspective, considering the effects of air resistance, gravity, and wind speed. The dynamic equations for the projectile were established and solved numerically. By analyzing the initial state of the artillery shell, all feasible solutions were determined for stationary targets. Subsequent discussions focused on estimating wind speeds for non-stationary targets, establishing empirical formulas that enable artillery units to achieve precise and rapid strikes without relying on computers or calculators.

When the target is fixed, using the scenario set in the first question with a horizontal range difference of 1200 meters between the target and artillery at the same elevation, the feasible solution ranges for θ and v are (3, 77) degrees and (147, 450) m/s, respectively. The minimum time required is 3.7 seconds, corresponding to v = 450 m/s and $\theta = 3°$.Then incorporate the effect of wind speed on air resistance. Investigate how the selection of initial angle is influenced by varying wind speeds at different heights and locations, while keeping the initial velocity fixed at 250 m/s. An empirical formula for estimation was fitted using the least squares method. The R-squared values are 0.99, 0.99, and 0.687, respectively, indicating excellent model fit.

After testing the empirical formula using randomly generated targets and wind speeds, the initial angle prediction deviated by approximately 0.05 degrees from the actual value under windless conditions. Under windy conditions, the initial angle prediction deviated by approximately 4 degrees from the actual value.

Key Words: Ballistics; Artillery; Launch Angle Estimation; Air Resistance; Wind Correction; Linear Fitting

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1 Introduction

1.1 Background

In early 19th-century warfare, artillery served as a pivotal force on the battlefield. Based on black powder propulsion, these cannons fired spherical iron projectiles typically weighing 5 kilograms with a diameter of 11 centimeters, achieving muzzle velocities up to 450 meters per second. Widely deployed 150 to 200 years ago, artillery officers' primary duty was rapidly determining firing parameters—namely muzzle velocity and elevation angle—based on target location and environmental factors like elevation and wind speed to ensure accurate projectile impact.

Ballistics is the science of projectile motion. In idealized models, the trajectory follows a standard parabola, with range primarily determined by the projectile's initial velocity and launch elevation angle. However, in real-world shooting scenarios, air resistance significantly impacts the motion of spherical projectiles. It not only causes velocity decay but also directly reduces range and lowers the apogee height. The magnitude of air resistance is primarily related to the projectile's flight speed, external shape, and atmospheric density.

This study aims to develop a practical method enabling artillery units to rapidly estimate the initial velocity and velocity angle of fired projectiles using empirical formulas. This approach avoids solving complex differential equations in dynamics, thereby significantly enhancing both the accuracy and speed of target engagement.



Figure 1. Military artillery of the Napoleonic era

1.2 Problem Restatement

Consider a cannon from 150 or 200 years ago that fires spherical projectiles with a mass of 5 kg and a diameter of 11 cm, achieving an initial velocity of up to 450 m/s. If the target is 1,200 meters away horizontally, and both the cannon and the target are located 500 meters above sea level, determine the initial velocity and elevation angle (relative to the horizontal plane) required to hit the target.

1.3 Problem Analyze

Regarding the first question, when the firing point and target are at the same elevation, the distance between the projectile and the target point is calculated using the Euclidean distance. If the distance between the two is less than 10 meters, the target is deemed to have been hit.

Regarding the second question, to enable artillery officers to rapidly estimate the muzzle velocity and elevation angle required to hit targets, we developed a model based on ballistic principles and linear approximation methods. Considering historical context, artillery muzzle velocity was typically controlled by charge quantity, allowing for two modes: half-charge (velocity approx. 250 m/s) and full-charge (velocity approx. 450 m/s). Since a velocity of 450 m/s requires longer time to strike targets at different horizontal elevations, the half-charge mode at 250 m/s is employed. Elevation angles are mechanically adjusted via the gun mount within a range of 0° to 45°. While precise calculation of ballistic trajectories affected by air resistance is complex, we observed an approximate linear relationship between elevation angle and either range or elevation difference within a certain range. Therefore, linear fitting is used for estimation.

2 Assumptions and Notations

2.1 Assumptions

- Assume the projectile is solid with uniform density distribution.
- Assume the projectile is perfectly smooth.
- Assume the air is uniform and remains at standard temperature and pressure during flight, as air density varies with temperature and pressure.
- Neglect the effect of supersonic speeds on the drag coefficient.
- Neglect the projectile's spin phenomenon.

2.2 Notations

Table 1: Main Notations for Artillery Projectile

Symbol	Definition	Numerical Value	Unit
h	Altitude	500	m
$v_{\max}$	Maximum projectile velocity	450	m/s
ρ	Air density	1.1679	kg/m^3
g	Gravitational acceleration	9.81	m/s^2
m	Projectile mass	5	kg
d	Projectile diameter	0.11	m
C_D	Drag coefficient	0.47	-
S_D	Maximum projectile cross-sectional area	0.0095	m^2
θ	Elevation angle	$[0°, 90°]$	°

3 Modeling and Analysis

3.1 Solving Ballistic Parameters for Fixed Targets

3.1.1 Air resistance

In the dynamic analysis of this model, besides the effect of gravity, we must also account for the interaction between the projectile and air during flight. To systematically elucidate the mechanism by which air resistance influences the trajectory, we first derive the governing principles of air resistance, then integrate these with gravitational effects to perform a comprehensive ballistic analysis.

The relationship between air resistance and flight speed can be analyzed using a simplified model. We consider a disk as the sphere's maximum cross-section, with the disk perpendicular to the direction of flight. The schematic is as follows:

Consider the airflow distribution on both sides of the disk. The airflow on the right side is not blocked by the sphere, with a relative velocity of $v_1 = v$ and gas pressure p_1; the airflow on the left side is blocked by the sphere, with a relative velocity $v_2 = 0$ and gas pressure p_2. According to Bernoulli's principle, we can derive the following equation:

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2 \tag{1}$$

Substituting $v_1 = v$ and $v_2 = 0$, we get:

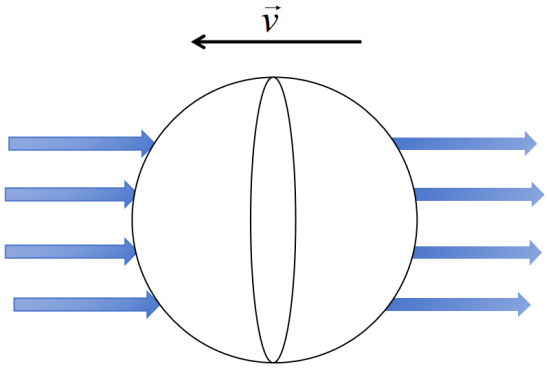


Figure 2. Airflow on both sides of the disk

$$p_2 - p_1 = \frac{1}{2}\rho v^2 \tag{2}$$

where ρ is the air density, and v is the relative velocity of the projectile with respect to the air.The effective area under pressure is $S_D = \pi r^2$; thus, the drag force acting on the disk is:

$$D = (p_1 - p_2)S_D = \frac{1}{2}\rho v^2 C_D S_D \tag{3}$$

This is a simplified expression for drag. The actual situation is more complex because the sphere is not disc-shaped, and air friction occurs with the sphere's surface. To correct the equation, we introduce the air resistance coefficient C_D. According to relevant literature [3], this coefficient is set to 0.47. The final equation is expressed in vector form:

$$\vec{D} = -0.235\pi r^2 \rho_0 \vec{v}^2 \tag{4}$$

3.1.2 Dynamic Equations

After determining the air resistance, we perform a force analysis on the projectile.

Since vector equations can be directly decomposed into two independent scalar equations in two-dimensional space, this makes their solution feasible. Therefore, the resultant external force is decomposed into the tangential and normal components in the Cartesian coordinate system.

In the tangential direction(τ-direction), the projectile is subjected only to air resistance and the tangential component of gravity. The direction of resistance opposes the direction of motion, and its magnitude is proportional to the square of the velocity. Therefore, its equation of motion is:

$$m\frac{dv}{dt} = -mg\sin\theta - D \tag{5}$$

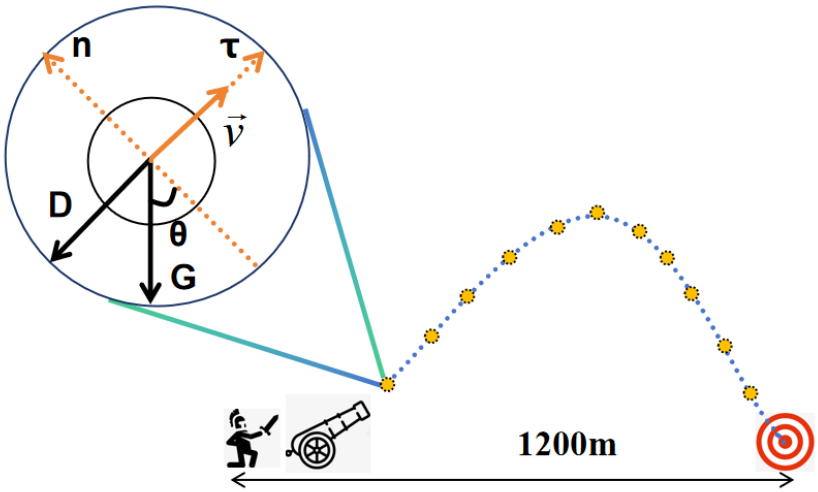


Figure 3. Force Analysis of Artillery Shells and Ballistic Trajectory

Normal direction (n-direction): In the normal direction, the projectile is subject only to the normal component of gravitational force. Therefore, its equation of motion is:

$$mv \cdot \frac{d\theta}{dt} = -mg\cos\theta \tag{6}$$

The rate of displacement change for a projectile is determined by the projections of its velocity vector onto each coordinate axis. In a Cartesian coordinate system, the velocity vector $\mathbf{v}$ can be decomposed into a horizontal component $v_x = v\cos\theta$ and a vertical component $v_y = v\sin\theta$, thereby establishing the differential equation for displacement:

$$\begin{cases} \frac{dx}{dt} = v\cos\theta \\ \frac{dy}{dt} = v\sin\theta \end{cases} \tag{7}$$

Combining equations (5)(6)(7), we obtain the following system of equations, while simultaneously specifying the initial conditions.

$$\begin{cases} \frac{dx}{dt} = v\cos\theta, \quad x(0) = 0 \\ \frac{dy}{dt} = v\sin\theta, \quad y(0) = 0 \\ \frac{dv}{dt} = -g\sin\theta - \frac{D}{m}, \quad v(0) = v_0 \\ \frac{d\theta}{dt} = -\frac{g\cos\theta}{v}, \quad \theta(0) = \theta_0 \end{cases} \tag{8}$$

3.1.3 Solution to the dynamic equation

The motion of the projectile is described by four coupled first-order ordinary differential equations:

$$\frac{d\mathbf{X}}{dt} = \mathbf{f}(t, \mathbf{X}), \quad \mathbf{X} = [x, y, \theta, v]^T \tag{9}$$

Due to the nonlinear nature of the air resistance term $D(v) \propto v^2$, this equation has no analytical solution. Given the initial conditions $\mathbf{X}_0 = [0, 0, \theta_0, v_0]^T$, the complete spatiotemporal trajectory can be solved using a fourth-order Runge-Kutta (RK4) numerical method.

Due to the strong nonlinear nature of the ballistic equation, the impact problem may yield multiple solutions (different angle-velocity combinations can all hit the target), and the distribution pattern of these solutions is unknown. To ensure all feasible solutions are found, we employ a full parameter space traversal strategy.

Subject to the problem constraints, we will discretize the continuous parameter space:

$$\begin{cases} \theta \in [0°, 90°], \quad \Delta\theta = 1° \\ v \in [0, 450] \text{ m/s}, \quad \Delta v = 1 \text{ m/s} \end{cases} \tag{10}$$

For each parameter combination (θ, v), we simulate the trajectory until one of the following conditions is met.

Hit Condition Considering the projectile's lethal radius and measurement error, we define the impact zone as a circular area centered at the target coordinates (x_t, y_t) with a radius of $r_{tol} = 10$ meters**:**

$$r = \sqrt{(x - x_t)^2 + (y - y_t)^2} \leq r_{tol} \tag{11}$$

Termination Conditions The simulation is terminated if any of the following occurs before a hit.

- Projectile hits the ground: $y < 0$
- Velocity becomes non-physical: $v \leq 0$
- Maximum simulation time exceeded: $t > 50$ s

The algorithm records all parameter sets that satisfy the hit condition for subsequent analysis.

Our iteration strategy is as follows: First, fix the launch angle and iterate through initial velocities to observe projectile trajectories; then, fix the initial velocity and iterate through launch angles.

3.1.4 Result Display

Figure 5 illustrates all launch angle and initial velocity combinations capable of hitting the target under air resistance conditions. Noticeable U-shaped distribution of hit parameters is observed on the launch angle-initial velocity plane. When the launch angle is small (less than 10°) or large (greater than 70°), the projectile requires a high

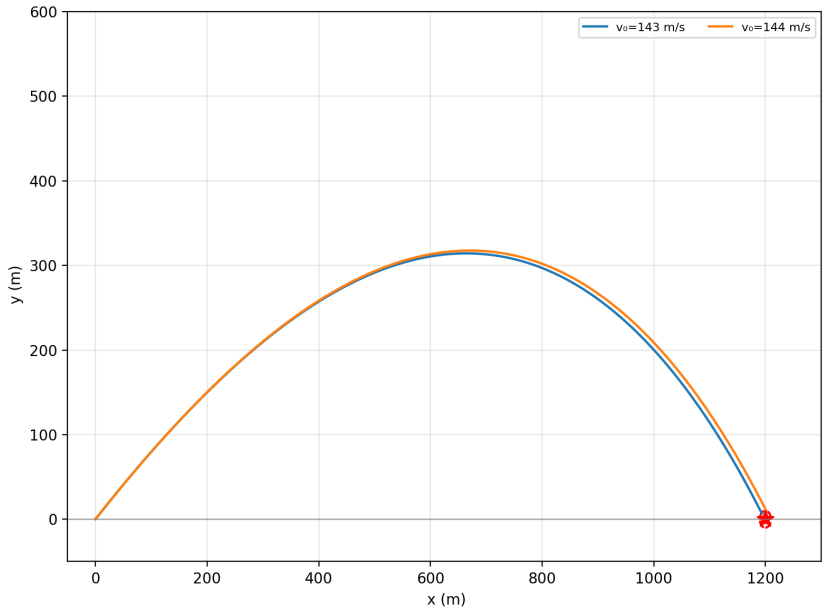


Figure 4. The trajectory of a projectile when θ is fixed at $40°$

initial velocity (approximately 350 – 450 m/s) to hit the target. Conversely, within the medium-angle range (approximately $30° - 45°$), a lower initial velocity (approximately 140–160 m/s) suffices to reach the target, indicating the most efficient energy utilization within this interval. Colors represent flight time, which increases significantly with greater launch angles—low-angle shots take approximately 3–6 seconds, while high-angle shots can exceed 40 seconds. Through iteration, we obtained 366 sets of feasible solutions. Figure 6 displays the spatial distribution of all impact trajectories. It is evident that as the launch angle increases, the apogee rises and the flight time extends; low-angle trajectories are flatter and achieve greater ranges. The overall distribution is continuous and smooth, indicating that the RK4 numerical integration solution is stable and reliable, accurately reflecting the true ballistic shape under aerodynamic drag.

In summary, as long as the combination of launch angle and initial velocity falls within the feasible region shown in the figure, the projectile will hit the target. Since this feasible region is difficult to express precisely through analytical equations or inequalities, only its distribution characteristics and approximate range are provided here: $v \in (143, 450)$ m/s, $\theta \in (3°, 77°)$. Specific parameter values can be obtained from the tables and code in the appendix.

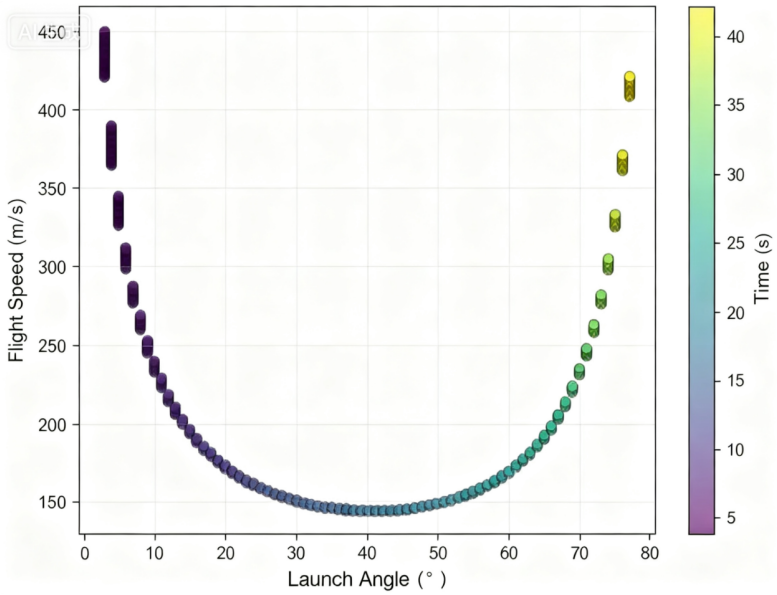


Figure 5. Hit Parameter Distribution

3.2 Determination of Ballistic Parameters for Moving Targets

3.2.1 Measurement under windless conditions

Traversing the entire parameter space (v_0, θ, R, H) significantly increases the computational complexity of the model. Considering practical conditions (where the initial velocity is controlled by the charge quantity) and the need for model simplification, only two common modes of initial velocity are considered: half charge (approximately 250 m/s) and full charge (approximately 450 m/s). Analysis indicates that an initial velocity of 450 m/s cannot cover the 1000–1500 m target range, and the time taken for the projectile to hit the target is excessively long when θ (45°, 90°). Therefore, this problem uniformly sets the initial velocity to 250 m/s and traverses the angle θ within the range (0°, 45°).

The determination of the estimation method involves two steps: First, fix the horizontal distance and observe the corresponding landing point heights for different θ values to establish the relationship between vertical height and angle. Second, fix the vertical height and observe the corresponding horizontal distances of landing points for different θ values to determine the relationship between horizontal position and angle. Finally, by integrating the results of numerical optimization, we propose a formulaic estimation method.

Case1:Elevation angle θ and horizontal distance x The graph of the relationship between θ and the horizontal distance x reveals a significant linear correlation between the two variables. Based on this, the least squares method is employed to fit the

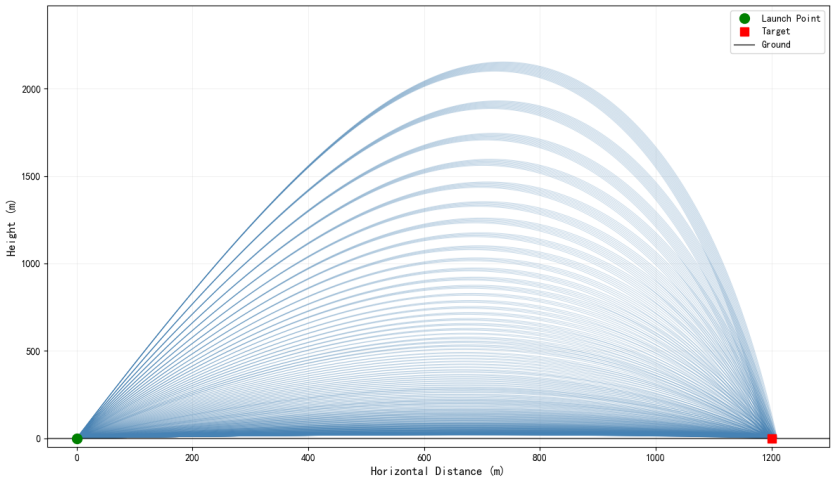


Figure 6. Shell Trajectory Diagram

relationship, with the model defined as $\theta = k_1 x + b_1$. Subsequently, the undetermined coefficients k_1 and b_1 within the model are determined.

Case2:Elevation angle θ and vertical height y At the same time, we also observe that θ exhibits a strong linear relationship with vertical height H in terms of the function, defined as $\theta = k_2 H + b_2$.

Result Analyze Through calculation, we obtain the coefficients.

$$\begin{cases} k_1 = 0.047514 \\ b_1 = 8.834154 \end{cases} \qquad \begin{cases} k_2 = 0.012216 \\ b_2 = -5.826506 \end{cases}$$

At the same time, we also calculated the respective decisive coefficients based on equation (12).

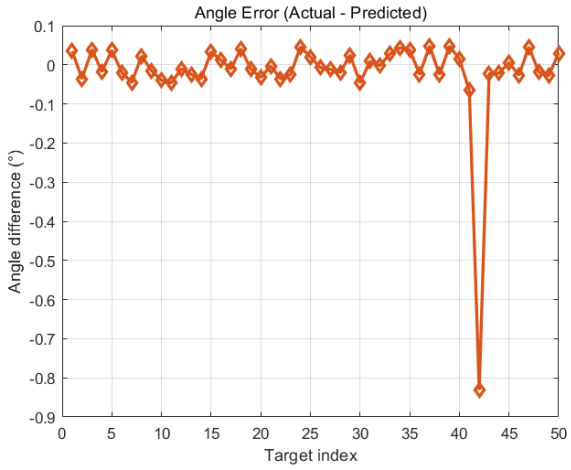
$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(\theta_i^{obs} - \theta_i^{pred}\right)^2}{\sum_{i=1}^{n} \left(\theta_i^{obs} - \bar{\theta}\right)^2} \tag{12}$$

Calculations yield $R_1^2 = 0.999$ and $R_2^2 = 0.996$, indicating excellent curve fitting performance.

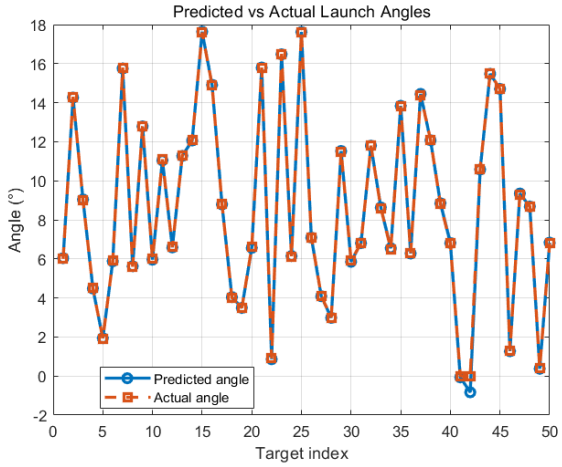
The reference point for the fitting equation is set at (1200, 0). Therefore, the calculated x and y values substituted into the equation represent the relative coordinates of the new target position relative to this reference point.

Under windless conditions, the relationship between horizontal position, height, and initial angle nearly strictly follows a linear distribution (with a coefficient of determination R^2 close to 1). Significance testing indicates the interaction term between horizontal position and height has negligible influence. Therefore, the x and y values can be substituted into their respective linear fitting equations, and the results summed to complete the calculation.

As shown in Figure 7, validation using randomly generated new targets demonstrates high agreement between predicted and actual values, further confirming the reliability of the aforementioned testing conclusions.



(a) Angle Error



(b) Predicted vs Actual Launch Angles

Figure 7. Launch Angle Prediction in Calm Wind Conditions

3.2.2 Wind Speed Correction

Wind conditions can be either headwind or tailwind. These two scenarios exert opposite effects on the distance to the shell's impact point. Headwind, where the wind direction opposes the projectile's trajectory, continuously impedes its forward motion, accelerating energy loss. This ultimately results in the projectile landing closer than it would have in calm conditions. Tailwind, where the wind direction aligns with the projectile's flight path, reduces energy dissipation, enabling the projectile to travel farther than it would under windless conditions.

The gunner adjusts the angle to compensate for range deviation, ensuring the shell overcomes wind effects and accurately hits the target. First, update the expression for air resistance.

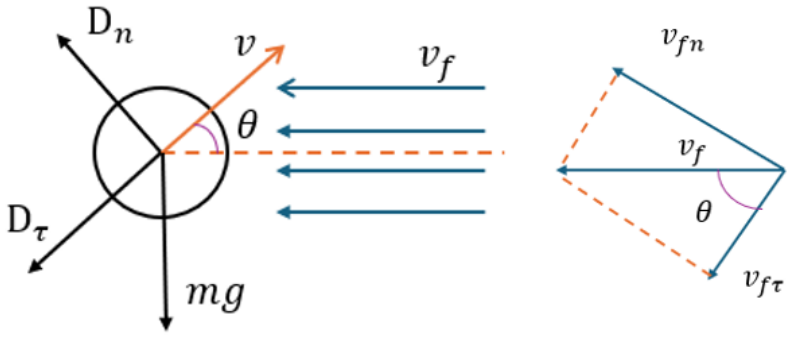


Figure 8. Analysis of Wind Speed Conditions

Decompose wind speed into tangential and normal components.

$$\begin{cases} v_{f\tau} = v_f \cos\theta \\ v_{fn} = v_f \sin\theta \end{cases} \tag{13}$$

Then calculate the relative velocity of the tangent and normal vectors relative to the projectile.

$$\begin{cases} v_{r\tau} = v - v_{f\tau} = v - v_f \cos\theta \\ v_{rn} = -v_f \sin\theta \\ \vec{v}_r = v_{r\tau} \cdot \vec{\tau} + v_{rn} \cdot \vec{n} \\ |\vec{v}_r| = \sqrt{v_{r\tau}^2 + v_{rn}^2} \end{cases} \tag{14}$$

Thus, we obtain the resistance update formula:

$$\begin{cases} D_\tau = -\frac{1}{2}\rho S_0 C_D |v_r| \cdot v_{r\tau} \\ D_n = -\frac{1}{2}\rho S_0 C_D |v_r| \cdot v_{rn} \end{cases} \tag{15}$$

Combining Equations (13), (14), and (15), the equation of motion is obtained as::

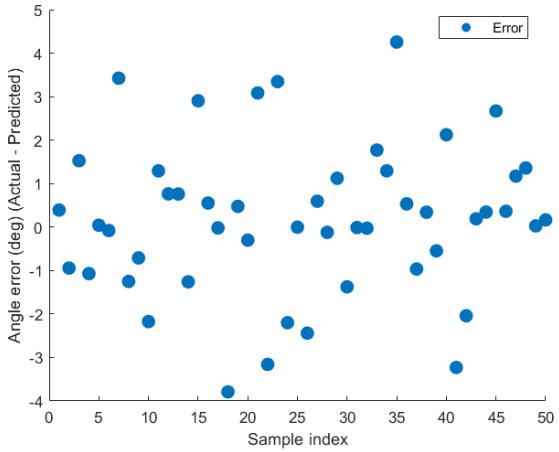
$$\begin{cases} \frac{dx}{dt} = v\cos\theta \\ \frac{dy}{dt} = v\sin\theta \\ \frac{dv}{dt} = -g\sin\theta - \frac{D_\tau}{m} \\ v \cdot \frac{d\theta}{dt} = -g\cos\theta + \frac{D_n}{m} \end{cases} \tag{16}$$

Set simulation benchmarks and parameter conditions: We define a wind speed axis where positive values represent headwinds and negative values represent tailwinds. The

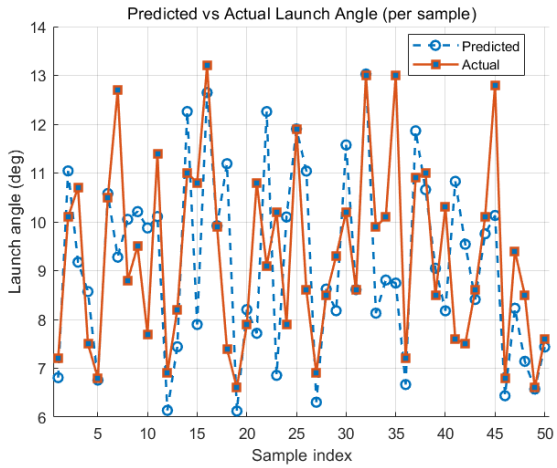
wind speed range is set to [-15, 15] (units: m/s). This range covers the vast majority of scenarios from strong tailwinds (-15 m/s, approximately 54 km/h) to strong headwinds (+15 m/s). Additionally, the initial angle range is [0, 45°].

Here, we also employ linear regression to fit the relationship between the emission angle θ and wind speed V_f. The fitted equation is defined as $\theta = k_3 V_f + b_3$. Solving this equation yields the regression coefficients: $k_3 = -0.049764$ and the constant term $b_3 = 8.762129$.

As shown in Figure 9, the model demonstrates excellent fitting performance in the beamwidth prediction task, exhibiting high prediction accuracy and minimal deviation.At the same time, the calculated coefficient of determination R_3^2 was 0.687.



(a) Angle Error



(b) Predicted vs Actual Launch Angles

Figure 9. Wind Speed Correction-Based Launch Angle Prediction

4 Strengths and Weaknesses

4.1 Strengths

1 We systematically addressed the issue. In developing the estimation method, we comprehensively considered multiple critical factors such as the target's horizontal position, elevation angle, and initial velocity selection, designing solutions tailored to the practical operational scenarios of artillery units.

2 Our model is highly practical. By establishing a linear empirical model linking elevation angle to horizontal range/altitude, artillery units can rapidly and effectively estimate firing parameters for targets within a horizontal range of 1000 to 1500 meters and relative to the artillery position at altitudes ranging from -178 to 778 meters—all without complex calculations on the battlefield.

3 Our solution demonstrates high feasibility. Using the method of controlled variables and least-squares fitting, we derived a universal relationship based on a typical scenario with a horizontal position of 1200 meters and a relative height of zero. Its applicability within adjacent ranges was validated through linear interpolation, confirming the robustness of the method.

4 Our assumptions and parameters align closely with historical context. The model is constructed based on given projectile parameters (mass 5kg, diameter 11cm) and the practicality of loading gunpowder into the cannon, specifying a typical initial velocity (250m/s). All assumptions conform to the technical conditions of the late 18th and early 19th centuries.

4.2 Weaknesses

1 The model's accuracy has inherent limitations. Despite our efforts to approximate the true ballistic trajectory through linear fitting, the non-linear relationship between wind speed and elevation angle, coupled with the interaction between wind speed and target position, results in certain model errors. The first shot may require a minor adjustment based on observation.

2 Certain environmental factors have been omitted. To ensure the model remains sufficiently simplified for rapid battlefield deployment, we have disregarded factors such as air turbulence and projectile spin. In practical applications, this may necessitate additional compensation by officers based on their experience.

3 The data sources are somewhat idealized. The model construction heavily relies on the specified projectile parameters and initial velocity. If batch variations exist in

the actual propellant charge, weight, or shape of the ammunition used, this may affect the accuracy of the model estimates.

5 Conclusion

After analyzing the first question, we identified all feasible solutions capable of hitting a stationary target: θ ranges from 3 to 77 degrees, and v ranges from 143 to 450 m/s. We then determined the trajectory that precisely hits the target among these feasible solutions. The solution with the shortest duration is selected as the optimal one: t = 3.7 s, v = 450 m/s, θ= 3°.

Building upon the first question, we designed a rapid aiming method for artillery based on a linear empirical model. The problem was first divided into windless and windy conditions. Using the control variable method, we obtained extensive data for the reference point (1200, 0). We then employed least-squares fitting to derive an empirical formula. After observing the target's relative position (x, y) relative to the reference point (1200, 0) and the approximate wind speed at that moment, artillery crews can substitute these values into the fitted empirical formula to calculate the required elevation angle for the gun.

In short, if the artillery adopts our strategy, it will be highly likely to strike the target quickly and accurately.

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A Appendix A: Numerical Code

Question 1:

```
clc;clear;close all;
a = pi/180;
global m g
m = 5; g = 9.81;
d = 0.11;
xm = 0; ym = 0; xt = 1200; yt = 0;
rho = 1.1679; S = pi*d^2/4; C = 0.47;
state_all = []; tout = [];
for thetam = 0:pi/180:pi/2
    for vm = 0:450
        state = [xm;ym;thetam;vm];
        state_all = [state_all,state];
    end
end
[~,n] = size(state_all);
stateout_all = {};
for i = 1:n
    xm = 0; ym = 0;
    state = state_all(:,i);
    vm = state(4);
    stateout = state;
    t0 = 0; dt = 0.01; iter = 0; t = 0;
    tf = 50;
```

```
            r = sqrt((xm-xt)^2+ym^2);
            while r > 10 && t < tf
                iter = iter + 1;
                r = sqrt((xm-xt)^2+ym^2);
                t = t0 + dt*iter;
                Q = 0.5*rho*S*C*vm^2;
                ke1 = statequation(state,t,Q);
                ke2 = statequation(state+0.5*ke1*dt,t+0.5*dt,Q);
                ke3 = statequation(state+0.5*ke2*dt,t+0.5*dt,Q);
                ke4 = statequation(state+ke3*dt,t+dt,Q);
                state = state + 1/6*(ke1+2*ke2+2*ke3+ke4)*dt;
                stateout = [stateout,state];
                % tout = [tout,t];
               vm = state(4); xm = state(1); ym = state(2);
                if vm < 0 || ym < -10
                   break;
                end
    end
            if r <= 10 && state(2) >=0
                stateout_all{end+1} = stateout;
                tout = [tout,t];
            end
        end
        function state = statequation(state,t,Q)
        global m g
        x = state(1);
        y = state(2);
        v = state(4);
        theta = state(3);
        xdot = v*cos(theta);
        ydot = v*sin(theta);
        vdot = -g*sin(theta)-Q/m;
        thetadot = -g*cos(theta)/v;
        state = [xdot;ydot;thetadot;vdot];
        end
```

Question 2:

```
    % clc;clear;close all;
    a = pi/180;
    global m g vf
```

```
m = 5; g = 9.81;
d = 0.11;
xm = 0; ym = 0; xt = 1200; yt = 0;
rho = 1.1679; S = pi*d^2/4; C = 0.47;
state_all = []; tout = [];
stateout_vf = {};
tout_vf = {};

vm = 250;
for thetam = 0:pi/1800:pi/2
state = [xm;ym;thetam;vm];
state_all = [state_all,state];
end
[~,n] = size(state_all);
stateout_all = {};
for vf = -15:15
stateout_all = {}; tout = [];
for i = 1:n
xm = 0; ym = 0;
state = state_all(:,i);
vm = state(4); thetam = state(3);
stateout = state;
t0 = 0; dt = 0.01; iter = 0; t = 0;
tf = 50;
r = sqrt((xm-xt)^2+ym^2);
while r > 10 && t < tf
iter = iter + 1;
r = sqrt((xm-xt)^2+ym^2);
t = t0 + dt*iter;
v_rel = sqrt((vm-vf*cos(thetam))^2 +(vf*sin(thetam))^2);
Q1 = 0.5*rho*S*C*v_rel*(vm-vf*cos(thetam));
Q2 = 0.5*rho*S*C*v_rel*(vf*sin(thetam));
ke1 = statequation1(state,t,Q1,Q2);
ke2 = statequation1(state+0.5*ke1*dt,t+0.5*dt,Q1,Q2);
ke3 = statequation1(state+0.5*ke2*dt,t+0.5*dt,Q1,Q2);
ke4 = statequation1(state+ke3*dt,t+dt,Q1,Q2);
state = state + 1/6*(ke1+2*ke2+2*ke3+ke4)*dt;
stateout = [stateout,state];
vm = state(4); xm = state(1); ym = state(2); thetam = state
```

```
        (3);
    if vm < 0 || ym < -10 || xm < -100
    break;
    end
    end
    if r <= 10
    stateout_all{end+1} = stateout;
    tout = [tout,t];
    end
    end
    stateout_vf{end+1} = stateout_all;
    tout_vf{end+1} = tout;
    end
    function state = statequation1(state,t,Q1,Q2)
    global m g vf
    x = state(1);
    y = state(2);
    v = state(4);
    theta = state(3);
    xdot = v*cos(theta);
    ydot = v*sin(theta);
    vdot = -g*sin(theta)-Q1/m;
    thetadot = -g*cos(theta)/v + Q2/(m*v);
    state = [xdot;ydot;thetadot;vdot];
    end
```